



Antonio Iacenda - FPGA Engineer

FPGA Engineer (MSc) in the defence industry, with hands-on experience in Xilinx FPGA design, VHDL RTL and SystemVerilog verification. Skilled in high-speed interfaces, custom and vendor IP integration, SoC design and hardware bring-up.

CONTACT

📍 Tivoli (Rome), Italy
☎ +39 377 080 0545
✉ tonioiacenda@gmail.com
🌐 tonio626.github.io
🌐 [LinkedIn: antonio-iacenda](#)

LANGUAGES

Italian: Native
English: Working Proficiency

TECH SKILLS

HDL & design

VHDL, SystemVerilog, RTL, FSM, pipelining, fixed-point design, CDC.

Video & links

Camera Link, CoaXPress 6.25 Gb/s, GTH transceivers, DRP, frame streaming.

On-chip interfaces

AXI4, AXI4-Lite master/slave, AXI4-Stream, FIFOs, BRAM, custom IP.

Serial & network

SPI/QSPI, I²C, UART, CRC-16, Ethernet L2, UDP, W5500.

SoC & embedded

Zynq UltraScale+ PS/PL, MicroBlaze, bare-metal C, STM32.

FPGA platforms

ZCU102, Cmod A7-35T (Artix-7).

Toolchain

Vivado, Vitis, Vitis HLS, IP Integrator/Packager, XSIM, Git.

Verification

SystemVerilog testbenches, assertions, XSIM behavioral and timing simulation, CDC, XDC.

Software & lab

Python, MATLAB, PSIM, Wireshark; oscilloscope, power supply, logic analyzer.

WORK EXPERIENCE

FPGA Engineer | Brain Technologies, assigned to MBDA

Rome, Italy

03/2026 - Present

- Design and integrate **VHDL RTL** for frame-grabber architectures and FPGA-based image-processing applications; work with high-speed Camera Link and CoaXPress interfaces, integrating custom logic and AXI4-Stream video datapaths with Host/Device IP cores.
- Validate FPGA designs on **ZCU102 evaluation boards** using FMC mezzanine connections and PL-connected SFP+ Ethernet interfaces, with on-chip probes and laboratory instruments.
- Configure **GTH transceivers via DRP** to switch the CoaXPress link rate at runtime from 3.125 to **6.25 Gb/s**, in accordance with the protocol specification.
- Integrate vendor IP with custom RTL through complex **FSMs, AXI4-Lite master/slave and AXI4-Stream** interfaces, following project ICDs and system requirements.
- Develop **SPI interfaces** between the FPGA and external peripherals; implement a custom high-level UART-based ASCII command/response protocol with **CRC-16/MODBUS**.
- Integrate and customize vendor and in-house IP using **Vivado IP Integrator and IP Packager**; interface programmable logic with the Zynq PS and develop bare-metal C applications in Vitis for board control and validation.
- Verify RTL with **SystemVerilog self-checking, assertion-based testbenches** and XSIM behavioral, post-synthesis and post-implementation simulations; analyze CDC, timing constraints and implementation results through synthesis and timing closure.
- Perform FPGA board bring-up and root-cause debugging using **ILA, VIO, mark_debug**, oscilloscopes and bench instrumentation; diagnose digital interfaces and data paths through on-chip captures, Python scripts and Wireshark.

Digital Hardware Engineer (Internship) | Sidereus Space Dynamics

Salerno, Italy

09/2025 - 02/2026

- Designed an avionics-oriented high-power synchronous DC-DC buck converter and its FPGA digital control chain (ADC acquisition, fixed-point controller, second-order delta-sigma modulation and DPWM).
- Modeled the power stage in MATLAB/PSIM; implemented SystemVerilog RTL on Artix-7, achieving about **1.25 µs acquisition-to-actuation latency** and validating control through STM32 hardware-in-the-loop tests.
- Performed functional/post-implementation simulation, resource and timing analysis, and laboratory validation with ILA and measurement instruments.

Curricular Internship | Microelectronics Laboratory

University of Salerno, Fisciano, Italy

03/2023 - 09/2023

- Designed, trained and compressed CNN models for TinyML using TensorFlow/TensorFlow Lite; deployed inference on STM32 microcontrollers.
- Developed C/C++ firmware with STM32CubeIDE/HAL and validated real-time sensor and inference behavior.

ENGINEERING FLOW

RTL & integration

Reusable VHDL modules, complex FSMs, vendor/custom IP and PS/PL interfaces.

Verification

Self-checking and assertion-based SystemVerilog testbenches; XSIM behavioral, post-synthesis and post-implementation simulations.

Implementation

Synthesis, place and route, XDC constraints, CDC analysis, resource review and timing closure.

Validation on hardware

FPGA evaluation-board tests; ILA/VIO and mark_debug captures, oscilloscope and bench measurements, Python scripts and Wireshark.

TRAINING

Advanced Digital Hardware Design course - Phil Salmony
FPGA/SoC board design, PDN, controlled impedance, DDR3, Gigabit Ethernet, USB 2.0 HS, eMMC.

EDUCATION

MSc Electronic Engineering (LM-29) - 110/110 cum laude

University of Salerno, Italy 2023 - 2025 (graduated 15/12/2025)

- **Thesis:** Design and Implementation of an FPGA-Based Digital Control System for a High-Power DC-DC Buck Converter.
- **Competencies developed:** Control Systems, Analog Electronics, Digital Electronics, Power Electronics, Signal Processing, Power Converters, measurement systems, electromagnetic compatibility (EMC), IoT systems (e.g., MQTT), and computer-vision-based systems.

BSc Electronic Engineering - 110/110 cum laude

University of Salerno, Italy 2020 - 2023

- Thesis: satellite-based territorial monitoring using a neural network and TinyML.

PROJECT HIGHLIGHTS

FPGA MNIST CNN accelerator and web demo - [GitHub repository](#)

- Trained and quantized a CNN in TensorFlow/Keras; built a **Vitis HLS accelerator** and integrated it as an IP in a Vivado block design targeting Cmod A7-35T.
- Linked MicroBlaze bare-metal C firmware, shared BRAM, AXI peripherals and an external **W5500 Ethernet controller over SPI**; exchanged 14 x 14 image frames and predictions over UDP.
- Built a Python/Flask browser demo for drawing digits and displaying FPGA classifications; verified the end-to-end path through UART debug and hardware probes.

32-bit FPGA microprocessor with FP32 ALU | SystemVerilog

- Designed a custom ISA, dual-port fetch RAM, floating-point adder/subtractor, iterative Booth multiplier and CORDIC divider; verified by simulation and FPGA implementation.

Custom STM32 PCB | Altium Designer

- Designed schematic and PCB layout with USB, 5 V-to-3.3 V regulation, SWD/Tag-Connect and an I²C IMU interface.

IoT environmental monitoring | Villa Regina, Boscoreale

- Integrated Raspberry Pi sensors, camera and LiDAR; ThingsBoard/MQTT telemetry, remote VPN access and KNIME analytics.

Coin recognition | MATLAB

- Implemented perspective calibration, segmentation and diameter-based classification with an interactive GUIDE application.